

Network Analysis Via PCAP File

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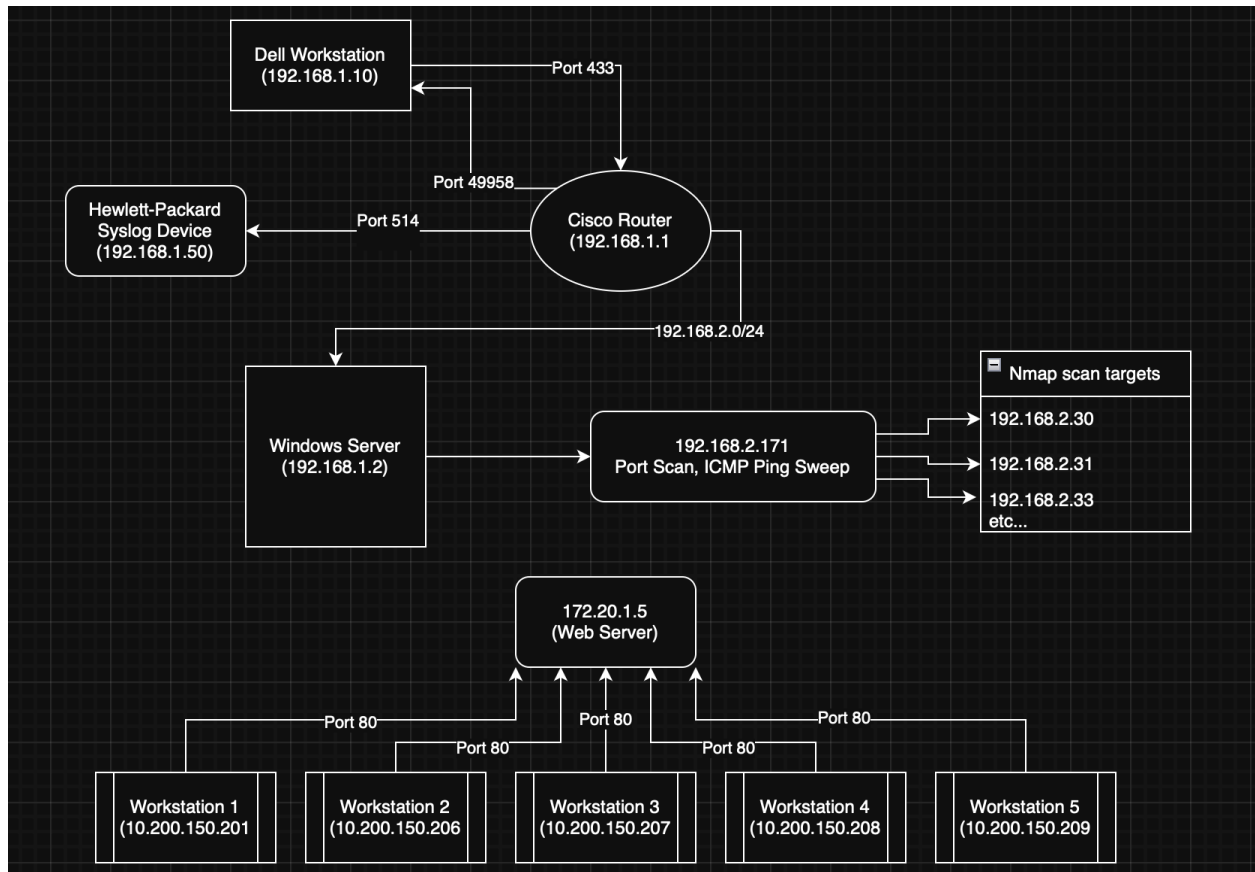
BLUF

The .pcap file I have received to analyze is of a network which consists of three main subnets. 192.168.1.0/24, 192.168.2.0/24, and 10.200.150.0/24. These subnets are interlinked through a Cisco network device at 192.168.1.1. Under the 192.168.1.0/24 subnet is a Hewlett-Packard (HP) server which receives the system logs of the Cisco network device, as well as a Dell workstation whose only connection is to the Cisco network device. There are five internal workstations on the 10.200.150.0/24 subnet, as well as a web server that they are all connected to at 170.20.1.5.

The most significant finding in this analysis was IP address 192.168.2.171. This address was conducting a full subnet TCP port scan across 192.168.2.0/24. This scan began with DNS lookups trying to resolve hostnames, followed by TCP port scanning, and concluded with ICMP echo requests. This behavior is most likely an automated network discovery tool such as nmap. It is not possible to determine if this scan was done maliciously or by a network administrator from the packet capture alone, but this scan is a very clear indicator of compromise and further investigation is required.

The last significant finding was the Windows Server at 192.168.1.2. It demonstrated clear server behavior such as listening on SMB (445), responding to ICMP echo requests, and receiving many more inbound connections than it initiates. There is another server at 172.20.1.2 which is used as an internal web server, operating on HTTP (80), communicating exclusively with the 10.200.150.x workstations. The overall network appears to be segmented, with the primary area of concern being the port scanning behavior from 192.168.2.171.

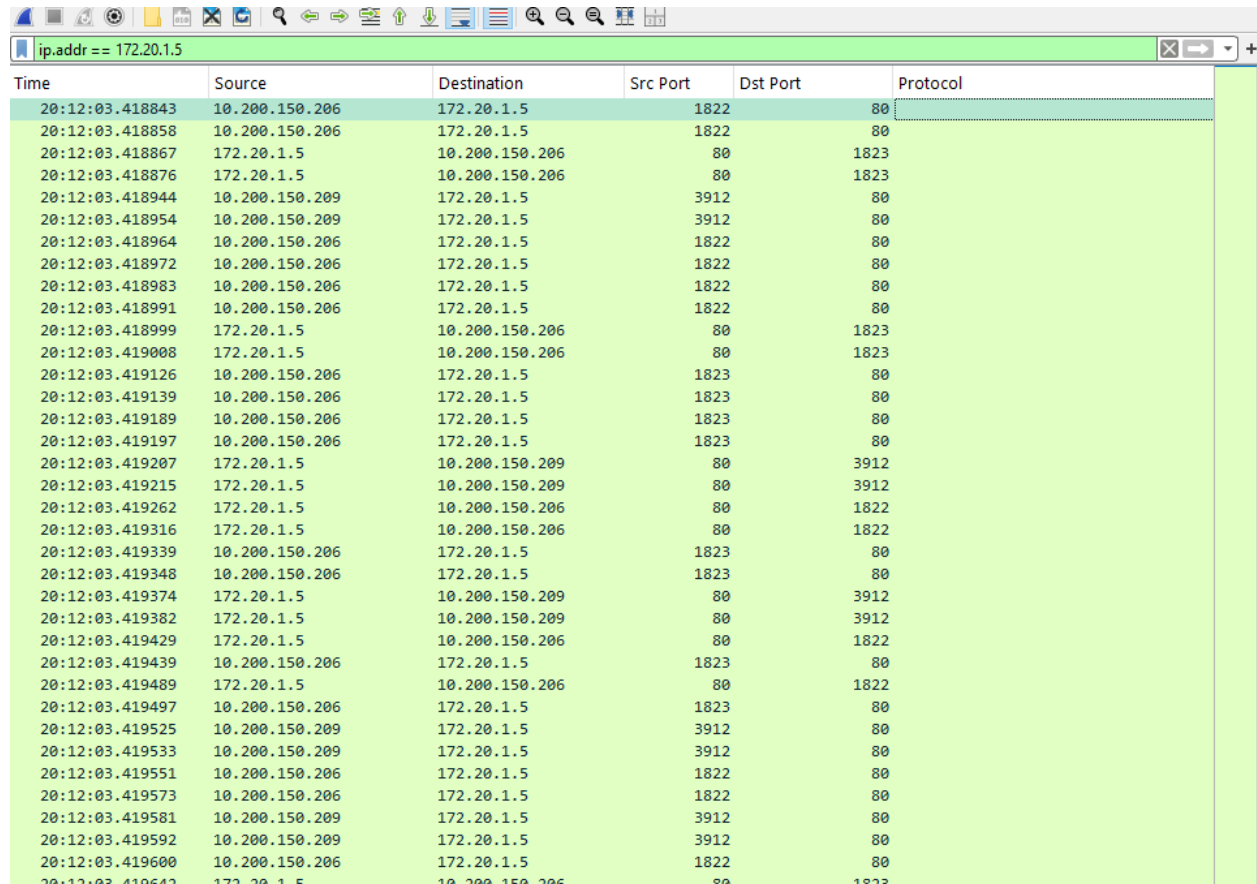
A network diagram that expresses your understanding of this network.



A list of hosts referenced by the PCAP.

172.20.1.5

Fig. 1 - IP address 172.20.1.5, operating only on port 80 (HTTP)



Time	Source	Destination	Src Port	Dst Port	Protocol
20:12:03.418843	10.200.150.206	172.20.1.5	1822	80	
20:12:03.418858	10.200.150.206	172.20.1.5	1822	80	
20:12:03.418867	172.20.1.5	10.200.150.206	80	1823	
20:12:03.418876	172.20.1.5	10.200.150.206	80	1823	
20:12:03.418944	10.200.150.209	172.20.1.5	3912	80	
20:12:03.418954	10.200.150.209	172.20.1.5	3912	80	
20:12:03.418964	10.200.150.206	172.20.1.5	1822	80	
20:12:03.418972	10.200.150.206	172.20.1.5	1822	80	
20:12:03.418983	10.200.150.206	172.20.1.5	1822	80	
20:12:03.418991	10.200.150.206	172.20.1.5	1822	80	
20:12:03.418999	172.20.1.5	10.200.150.206	80	1823	
20:12:03.419008	172.20.1.5	10.200.150.206	80	1823	
20:12:03.419126	10.200.150.206	172.20.1.5	1823	80	
20:12:03.419139	10.200.150.206	172.20.1.5	1823	80	
20:12:03.419189	10.200.150.206	172.20.1.5	1823	80	
20:12:03.419197	10.200.150.206	172.20.1.5	1823	80	
20:12:03.419207	172.20.1.5	10.200.150.209	80	3912	
20:12:03.419215	172.20.1.5	10.200.150.209	80	3912	
20:12:03.419262	172.20.1.5	10.200.150.206	80	1822	
20:12:03.419316	172.20.1.5	10.200.150.206	80	1822	
20:12:03.419339	10.200.150.206	172.20.1.5	1823	80	
20:12:03.419348	10.200.150.206	172.20.1.5	1823	80	
20:12:03.419374	172.20.1.5	10.200.150.209	80	3912	
20:12:03.419382	172.20.1.5	10.200.150.209	80	3912	
20:12:03.419429	172.20.1.5	10.200.150.206	80	1822	
20:12:03.419439	10.200.150.206	172.20.1.5	1823	80	
20:12:03.419489	172.20.1.5	10.200.150.206	80	1822	
20:12:03.419497	10.200.150.206	172.20.1.5	1823	80	
20:12:03.419525	10.200.150.209	172.20.1.5	3912	80	
20:12:03.419533	10.200.150.209	172.20.1.5	3912	80	
20:12:03.419551	10.200.150.206	172.20.1.5	1822	80	
20:12:03.419573	10.200.150.206	172.20.1.5	1822	80	
20:12:03.419581	10.200.150.209	172.20.1.5	3912	80	
20:12:03.419592	10.200.150.209	172.20.1.5	3912	80	
20:12:03.419600	10.200.150.206	172.20.1.5	1822	80	
20:12:03.419642	172.20.1.5	10.200.150.206	80	1823	

This IP address, 172.20.1.5, is a web server running primarily on port 80 (HTTP). The use of port 80 may be concerning, as credentials and session tokens are transmitted in plaintext. Port 443 would be a better option for network traffic.

Fig. 2 - 172.20.1.5 communicates exclusively with 10.200.150.201, 206, 207, 208 & 209

Ethernet · 6		IPv4 · 5	IPv6	TCP	UDP				
Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	
10.200.150.201	172.20.1.5	112,100	14 MB	7	112,100	100.00%	79,384	5 MB	
10.200.150.206	172.20.1.5	113,370	14 MB	1	113,370	100.00%	81,378	5 MB	
10.200.150.208	172.20.1.5	90,813	11 MB	4	90,813	100.00%	65,212	4 MB	
10.200.150.209	172.20.1.5	114,733	15 MB	2	114,733	100.00%	81,595	5 MB	

10.200.150.201
10.200.150.206
10.200.150.207
10.200.150.208
10.200.150.209

These 5 internal IP addresses are internal workstations, and generate high volumes of traffic (around 1.1 to 1.4 million packets). They exclusively communicate with the web server at 172.20.1.5

Fig. 3 - 10.200.150.201 communicates exclusively with 172.20.1.5 (1.4M packets)

Ethernet · 2	IPv4 · 1	IPv6	TCP	UDP					
Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A
10.200.150.201	172.20.1.5	1,444,653	192 MB	7	1,444,653	100.00%	1,025,170	65 MB	419,483

Fig. 4 - 10.200.150.206 communicates exclusively with 172.20.1.5 (1.3M packets)

Ethernet · 2	IPv4 · 1	IPv6	TCP	UDP					
Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A
10.200.150.206	172.20.1.5	1,331,183	174 MB	1	1,331,183	100.00%	951,848	60 MB	379,335

Fig. 5 - 10.200.150.207 communicates exclusively with 172.20.1.5 (1.2M packets)

Ethernet · 2	IPv4 · 1	IPv6	TCP	UDP					
Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A
10.200.150.207	172.20.1.5	1,265,109	165 MB	5	1,265,109	100.00%	904,016	57 MB	361,093

Fig. 6 - 10.200.150.208 communicates exclusively with 172.20.1.5 (1.1M packets)

Ethernet · 2	IPv4 · 1	IPv6	TCP	UDP					
Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A
10.200.150.208	172.20.1.5	1,121,238	147 MB	4	1,121,238	100.00%	801,599	51 MB	319,639

Fig. 7 - 10.200.150.209 communicates exclusively with 172.20.1.5 (1.1M packets)

Ethernet · 2	IPv4 · 1	IPv6	TCP	UDP					
Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A
10.200.150.209	172.20.1.5	1,429,699	186 MB	2	1,429,699	100.00%	1,024,496	65 MB	405,203

192.168.1.1

Fig. 8 - 192.168.1.1 is under the “Cisco” vendor, and is a network device or HTTPS admin interface

```
▼ Ethernet II, Src: Cisco_fe:97:a0 (d0:d0:fd:fe:97:a0), Dst: Dell_55:36:af (00:24:e8:55:36:af)
  ▼ Destination: Dell_55:36:af (00:24:e8:55:36:af)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  ▼ Source: Cisco_fe:97:a0 (d0:d0:fd:fe:97:a0)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
```

This screenshot also shows that **192.168.1.10** is a Dell workstation and that it only accesses the Cisco web interface.

192.168.1.50

Fig. 9 - 192.168.1.50 is under the “Hewlett-Packard (HP)” vendor

```
▼ Ethernet II, Src: Cisco_fe:97:a0 (d0:d0:fd:fe:97:a0), Dst: HewlettPacka_6a:67:47 (00:16:35:6a:67:47)
  ▼ Destination: HewlettPacka_6a:67:47 (00:16:35:6a:67:47)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
```

Fig. 10 - The HP device at 192.168.1.50 operates only on port 514. Port 514 is used for Syslog (System Logging Protocol) over UDP. 192.168.1.50 only receives data; this device is most likely a central logging system for the Cisco network device.

```
▼ User Datagram Protocol, Src Port: 514, Dst Port: 514
  Source Port: 514
  Destination Port: 514
  Length: 197
```

192.168.1.2

This IP address is very likely a Windows Server. Indicators include: listening on port 445 (SMB), responding to pings from 192.168.2.x IP addresses, responding to host initiating SMB sessions, and having many more inbound connections than outbound.

Fig. 11 - Usage of port 445

192.168.2.64	2383	192.168.1.2	445	TCP	130	2383 → 445[Packet size limited during capt
192.168.1.2	445	192.168.2.64	2383	TCP	186	445 → 2383[Packet size limited during capt
192.168.1.2	445	192.168.2.64	2383	TCP	186	445 → 2383[Packet size limited during capt
192.168.1.2	445	192.168.2.64	2383	TCP	186	445 → 2383[Packet size limited during capt
192.168.1.2	445	192.168.2.64	2383	TCP	186	445 → 2383[Packet size limited during capt
192.168.2.64	2383	192.168.1.2	445	TCP	146	2383 → 445[Packet size limited during capt
192.168.2.64	2383	192.168.1.2	445	TCP	146	2383 → 445[Packet size limited during capt
192.168.2.64	2383	192.168.1.2	445	TCP	146	2383 → 445[Packet size limited during capt
192.168.1.2	445	192.168.2.64	2383	TCP	228	445 → 2383[Packet size limited during capt
192.168.1.2	445	192.168.2.64	2383	TCP	228	445 → 2383[Packet size limited during capt
192.168.1.2	445	192.168.2.64	2383	TCP	228	445 → 2383[Packet size limited during capt
192.168.2.64	2383	192.168.1.2	445	TCP	138	2383 → 445[Packet size limited during capt
192.168.2.64	2383	192.168.1.2	445	TCP	138	2383 → 445[Packet size limited during capt
192.168.2.64	2383	192.168.1.2	445	TCP	138	2383 → 445[Packet size limited during capt
192.168.1.2	445	192.168.2.64	2383	TCP	188	445 → 2383[Packet size limited during capt
192.168.1.2	445	192.168.2.64	2383	TCP	188	445 → 2383[Packet size limited during capt
192.168.1.2	445	192.168.2.64	2383	TCP	188	445 → 2383[Packet size limited during capt

Fig. 12 - Responding to 192.168.2.x pings

192.168.2.36	192.168.1.2	ICMP	74	Echo (ping) request
192.168.2.36	192.168.1.2	ICMP	74	Echo (ping) request
192.168.2.36	192.168.1.2	ICMP	74	Echo (ping) request
192.168.2.36	192.168.1.2	ICMP	74	Echo (ping) request
192.168.1.2	192.168.2.36	ICMP	74	Echo (ping) reply
192.168.1.2	192.168.2.36	ICMP	74	Echo (ping) reply
192.168.1.2	192.168.2.36	ICMP	74	Echo (ping) reply
192.168.1.2	192.168.2.36	ICMP	74	Echo (ping) reply

192.168.2.171

This IP address is very likely a workstation within the 192.168.2.x subnet which is running a series of scans. I will show more evidence for this in the IOC section.

A list of port numbers that each of these hosts offers services from.

Core port numbers

445 - Windows File Sharing

- Files are shared over this port, lots of Windows management activity. Most traffic on this port is normal in Windows environments

53 - DNS

- Domain name resolution is incredibly common on networks.

80 - HTTP

- Unencrypted web traffic without TLS

443 - HTTPS

- Encrypted web traffic, very common

514 - Syslog

- Devices sending logs to a storage device

Additional port numbers (appeared in port scan)

9 - Discard Protocol

21 - FTP

22 - SSH

23 - Telnet

25 - SMTP

42 - WINS Replication

89 - OSPF (Open Shortest Path First)

113 - Ident

389 - LDAP

636 - LDAPS

3389 - RDP

Any Indicators of Compromise (IOCs) that you might identify, or that you think are potentially suspicious.

IP Address **192.168.2.171**

This IP address shows very clear signs of port scanning within the 192.168.2.x subnet.

Fig. 13 - 2.171 starts by trying to resolve hostnames (port 53, DNS).

[illegible]

Note: The Nmap command “nmap 192.168.2.0/24” was most likely used as it will resolve hostnames first by default.

Fig. 14 - 2.171 scanning every port in the 192.168.2.x subnet

[illegible]

While it is difficult to determine if this scan is done by a network administrator or is malicious in nature, 192.168.2.171 is performing a textbook Nmap scan. This is a potential indicator of compromise.